Châu Gia Khang

Fullstack Developer

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SUMMARY

Fullstack developer, strongest on backend (**TypeScript · NestJS · Golang**), with **~1 year** of professional experience plus a marketplace of my own running in production. Deepest on the messy parts of distributed systems - transactional outbox, idempotent consumers with dedup and DLQ, CQRS read projections, and OpenTelemetry tracing that survives an async boundary. On the frontend, build production **React/Next.js** with SSR, TanStack Query, and state management.

SKILLS

Languages: TypeScript, JavaScript, SQL, Golang
Backend: NestJS, Node.js, Express.js, REST APIs, WebSocket (Socket.IO), Passport.js (JWT, OAuth2), RBAC, Swagger/OpenAPI, Rate Limiting (Throttler)
Testing: Jest (unit), Supertest, Testcontainers (integration & E2E on ephemeral Postgres)
Data & Messaging: PostgreSQL, TypeORM, MongoDB, Redis, RabbitMQ, Full-text search (Postgres tsvector/pg_trgm), sqlc
DevOps: Docker, docker-compose, GitHub Actions (CI/CD), GHCR, Render, Vercel; CodeQL, Dependabot
Frontend: React, Next.js, Zustand, Shadcn/UI, TanStack Query, SEO (SSR, JSON-LD), Tailwind CSS
Patterns: Transactional Outbox, CQRS, Event-driven, Idempotency, Strangler Fig, Two-stage retrieval

EXPERIENCE

Backend Developer (Fresher) 03/2026 – 07/2026

- Developed backend features for an ERP scheduling platform for a tutoring center: tutor onboarding and account provisioning, class scheduling for academic coordinators and tutors, grading, quizzes, and an online learning module.
- Built REST APIs with NestJS and contributed UI with React + shadcn/ui in a team codebase (deployed on Kubernetes); worked in Agile sprints, clarifying requirements with business analysts and performing manual QC.

Fullstack Developer Intern 08/2025 – 01/2026

- Built the document layer for a U.S. notary office with Express.js and React: S3 file storage, per-document access control, and a 5-state document lifecycle with a per-step audit trail of who acted.

PROJECT

Multi-Vendor E-Commerce Platform - solo, self-directed *03*/2026 –present

Live: link · *Code:* link · *API docs (Swagger, 113 endpoints):* link

NestJS · Golang · PostgreSQL · TypeORM · RabbitMQ · Redis · Cloudinary · Next.js · Docker · GitHub Actions · Tailwind CSS

- Built a multi-vendor marketplace **(15 feature modules, 113 REST endpoints)**: multi-shop cart & checkout with idempotency keys, pessimistic locking for stock, distance-based shipping (geocoding + Haversine + live geocoding API - LocationIQ/Nominatim), coupons, reviews, returns, and seller payouts.
- Built a **Go Search Microservice** (own **Neon Postgres**, database-per-service): **full-text search** (tsvector + unaccent + GIN, ts_rank_cd ranking, sqlc type-safe queries) with **pg_trgm trigram** fallback for partial/misspelled queries, served via two-stage retrieval (index returns ranked IDs → monolith hydrates & applies volatile filters). Cut keyword-search **p95 6.07s → 186ms (~33×, 100 VU)**; EXPLAIN-verified Seq Scan → GIN bitmap scan.
- Extracted a **Standalone Notification Microservice** from the monolith (own **Supabase Postgres**, **database-per-service + CQRS read projection**): **14 domain event** types over a transactional outbox → RabbitMQ → idempotent consumer with dedup + DLQ, plus cross-process realtime (**Socket.IO + Redis**); strangler-fig migration with **zero-downtime cutover** verified in production.
- Implemented authentication with **Passport.js** (local login, **JWT access/refresh tokens**, **Google OAuth2**) enforced by role-based route guards; hardened the API with Helmet, rate limiting, strict DTO validation/whitelisting, httpOnly/secure cookies behind a CORS whitelist, and **Cloudflare Turnstile** on auth endpoints.
- Set up **CI/CD** (GitHub Actions → GHCR image → Render), **Docker** multi-stage, structured logging (pino), Sentry, CodeQL, **SEO** (SSR + JSON-LD); added **Jest** unit + **Testcontainers** integration tests.
- Instrumented **all three services** with **OpenTelemetry**: one request produces a single trace across the async outbox boundary and the process boundary (RabbitMQ headers) into the downstream notification and search services. Added RED metrics via **prom-client** with a **Grafana** dashboard provisioned as code.

EDUCATION

University of Information Technology (UIT), VNU-HCM 06/2025

GPA: 8.0 / 10 · TOEIC 765